Mini Project 2: Tight-binding model

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1 Problem Statement

The electronic structures of two (imaginary) two-dimensional crystals, A and B, are described by a tight-binding model with a two-orbital basis. The dependence of the overlap and hopping parameters, η_{nm} and t_{nm} on the distance R between lattice sites is given by the following functions:

$$\eta_{11}(R) = \left(1 + \frac{R}{d_0} + \frac{R^2}{3d_0^2}\right) e^{-\frac{R}{d_0}}$$

$$\eta_{12}(R) = \frac{1}{2}\left(1 + \frac{R}{d_0} + \frac{R^2}{64d_0^2}\right) e^{-\frac{R}{2d_0}} - \frac{1}{2}e^{-\frac{R}{d_0}}$$

$$\eta_{22}(R) = \left(1 + \frac{R}{2d_0} + \frac{R^2}{24d_0^2}\right) e^{-\frac{R}{2d_0}}$$

$$t_{11}(R) = -t_0\left(1 + \frac{2R}{d_0}\right) e^{-\frac{R}{d_0}}$$

$$t_{12}(R) = -t_0\left(1 + \frac{2R}{d_0}\right) e^{-\frac{R}{d_0}} + t_0\left(1 + \frac{R}{16d_0}\right) e^{-\frac{R}{2d_0}}$$

$$t_{22}(R) = -t_0\left(1 + \frac{R}{2d_0} - \frac{R^2}{24d_0^2}\right) e^{-\frac{R}{2d_0}}$$

where d_0 and t_0 are positive constants that we use as length and energy units, respectively. The expressions for the hopping parameters are valid for $R > 3d_0$ and we have $t_{nm}(0) = 0$. The atomic energies associated with the two orbitals are $\epsilon_1 = 0$ and $\epsilon_2 = 0.8t_0$.

1. Calculate the electronic band structures of the two crystals assuming that the lattice constant is: (i) $a = 5d_0$; (ii) $a = 7d_0$; (iii) $a = 9d_0$. Include the nearest-neighbor and the next-nearest-neighbor contributions.
2. Compare the results in (1) with the corresponding energy spectra calculated: (i) using only nearest neighbor contributions; (ii) neglecting the overlap integral, $\eta_{nm}(R) = \delta_{n,m}\delta_{R,0}$.

Note. The two crystals considered in this project have the following two-dimensional structures:

1. **Crystal A:** A square lattice with lattice constant a and primitive lattice vectors

$$\mathbf{a}_1 = (a, 0), \qquad \mathbf{a}_2 = (0, a).$$

   The primitive unit cell contains one lattice site at $\boldsymbol{\tau}_1 = (0, 0)$, with two orbitals labeled 1 and 2.

2. **Crystal B:** A rectangular lattice with lattice constants a and $b = 1.5a$, and primitive lattice vectors

$$\mathbf{a}_1 = (a, 0), \qquad \mathbf{a}_2 = (0, b).$$

   The primitive unit cell contains a two-site basis:

   - **Site 1:** $\boldsymbol{\tau}_1 = (0, 0)$, with orbitals 1 and 2.
   - **Site 2:** $\boldsymbol{\tau}_2 = (a/2, b/2)$, with orbital 1 only.

To skip the derivation and see the results directly, see pages 4 and 7 for Question 1 (Figures 1 and 3), and pages 5 and 8 for Question 2 (Figures 2 and 4).

2 Electronic Band Structure

2.1 Crystal A

Using the prescribed hopping (t_{nm}) and overlap (η_{nm}) matrix elements, the electronic band structure is obtained by solving the generalized tight-binding eigenvalue problem:

$$[\bar{\epsilon}\bar{\eta}(\mathbf{k}) + \bar{t}(\mathbf{k})]\, \mathbf{c}_\nu(\mathbf{k}) = E_\nu(\mathbf{k})\bar{\eta}(\mathbf{k})\mathbf{c}_\nu(\mathbf{k}), \tag{1}$$

where $\bar{\epsilon}$, $\bar{\eta}(\mathbf{k})$, and $\bar{t}(\mathbf{k})$ denote the on-site energy, overlap, and hopping matrices, respectively, while $E_\nu(\mathbf{k})$ represents the energy of band ν. Crystal A consists of a square lattice with two atomic orbitals per site, resulting in two electronic bands. The on-site energy matrix is

$$\bar{\epsilon} = \begin{pmatrix} \epsilon_1 & 0 \\ 0 & \epsilon_2 \end{pmatrix}, \tag{2}$$

with $\epsilon_1 = 0$ and $\epsilon_2 = 0.8t_0$. Including first- and second-nearest-neighbor interactions, the overlap and hopping matrices can be expressed as

$$\bar{\eta}(\mathbf{k}) = \begin{pmatrix} 1 + \eta_{11}(a)f_{\mathrm{NN}} + \eta_{11}(\sqrt{2}a)f_{\mathrm{NNN}} & \eta_{12}(a)f_{\mathrm{NN}} + \eta_{12}(\sqrt{2}a)f_{\mathrm{NNN}} \\ \eta_{21}(a)f_{\mathrm{NN}} + \eta_{21}(\sqrt{2}a)f_{\mathrm{NNN}} & 1 + \eta_{22}(a)f_{\mathrm{NN}} + \eta_{22}(\sqrt{2}a)f_{\mathrm{NNN}} \end{pmatrix}, \tag{3}$$

and

$$\bar{t}(\mathbf{k}) = \begin{pmatrix} t_{11}(a)f_{\mathrm{NN}} + t_{11}(\sqrt{2}a)f_{\mathrm{NNN}} & t_{12}(a)f_{\mathrm{NN}} + t_{12}(\sqrt{2}a)f_{\mathrm{NNN}} \\ t_{21}(a)f_{\mathrm{NN}} + t_{21}(\sqrt{2}a)f_{\mathrm{NNN}} & t_{22}(a)f_{\mathrm{NN}} + t_{22}(\sqrt{2}a)f_{\mathrm{NNN}} \end{pmatrix}, \tag{4}$$

where the nearest-neighbor (NN) and next-nearest-neighbor (NNN) structure factors are

$$f_{\mathrm{NN}}(\mathbf{k}) = 2\left[\cos(k_x a) + \cos(k_y a)\right], \tag{5}$$

$$f_{\mathrm{NNN}}(\mathbf{k}) = 4\cos(k_x a)\cos(k_y a). \tag{6}$$

The real-space square lattice and its corresponding reciprocal lattice are illustrated in Figs. 1(a) and (b), respectively. The electronic band structures were calculated along the high-symmetry path $\Gamma - X - M - \Gamma$ for three lattice constants: $a = 5d_0$, $7d_0$, and $9d_0$.

For $a = 5d_0$ [Fig. 1(c)], the calculated bands exhibit pronounced divergences. Furthermore, in Fig. 2 (left), both the NN + NNN and NN approximations show divergences near the X and M points. These features are associated with the singularity of the overlap matrix $\bar{\eta}(\mathbf{k})$, indicating a breakdown of the generalized eigenvalue description at this lattice spacing. Consequently, the divergent eigenvalues should not be interpreted as physically meaningful electronic energies within the present model.

Increasing the lattice constant to $a = 7d_0$ [Fig. 1(d)] produces a more regular band structure, with a clear separation between the two bands. However, divergences are still present in the nearest-neighbor approximation [Fig. 2 (middle)].

At $a = 9d_0$ [Fig. 1(e)], the bands become progressively flatter, while their separation approaches the difference between the isolated-orbital energies. At this length scale, all three approximations approach convergence rapidly due to the exponential decay of the local orbital contributions in this limit, causing a clear narrowing of the bands as the lattice parameter a increases as expected. As the lattice expands, inter-site coupling weakens, and the $\mathbf{k}$-dependence of the eigenvalues is reduced. In the limit of large interatomic separation, $\bar{t}(\mathbf{k}) \to 0$ and $\bar{\eta}(\mathbf{k}) \to \mathbf{I}$, so the two bands approach the isolated-orbital energies $\epsilon_1 = 0$ and $\epsilon_2 = 0.8t_0$. The corresponding reduction in bandwidth reflects the increasing localization of the electronic states.

3 Crystal B

For crystal B, the same tight-binding procedure is applied, but the construction of the matrices requires additional care because the primitive cell contains two atomic sites. The first site hosts two orbitals, while the second hosts only one, resulting in three electronic bands. Using the orbital basis $(1,1)$, $(1,2)$, and $(2,1)$, where the first index identifies the atomic site and the second the orbital, the on-site energy matrix becomes

$$\bar{\epsilon} = \begin{pmatrix} \epsilon_1 & 0 & 0 \\ 0 & \epsilon_2 & 0 \\ 0 & 0 & \epsilon_1 \end{pmatrix}. \tag{7}$$

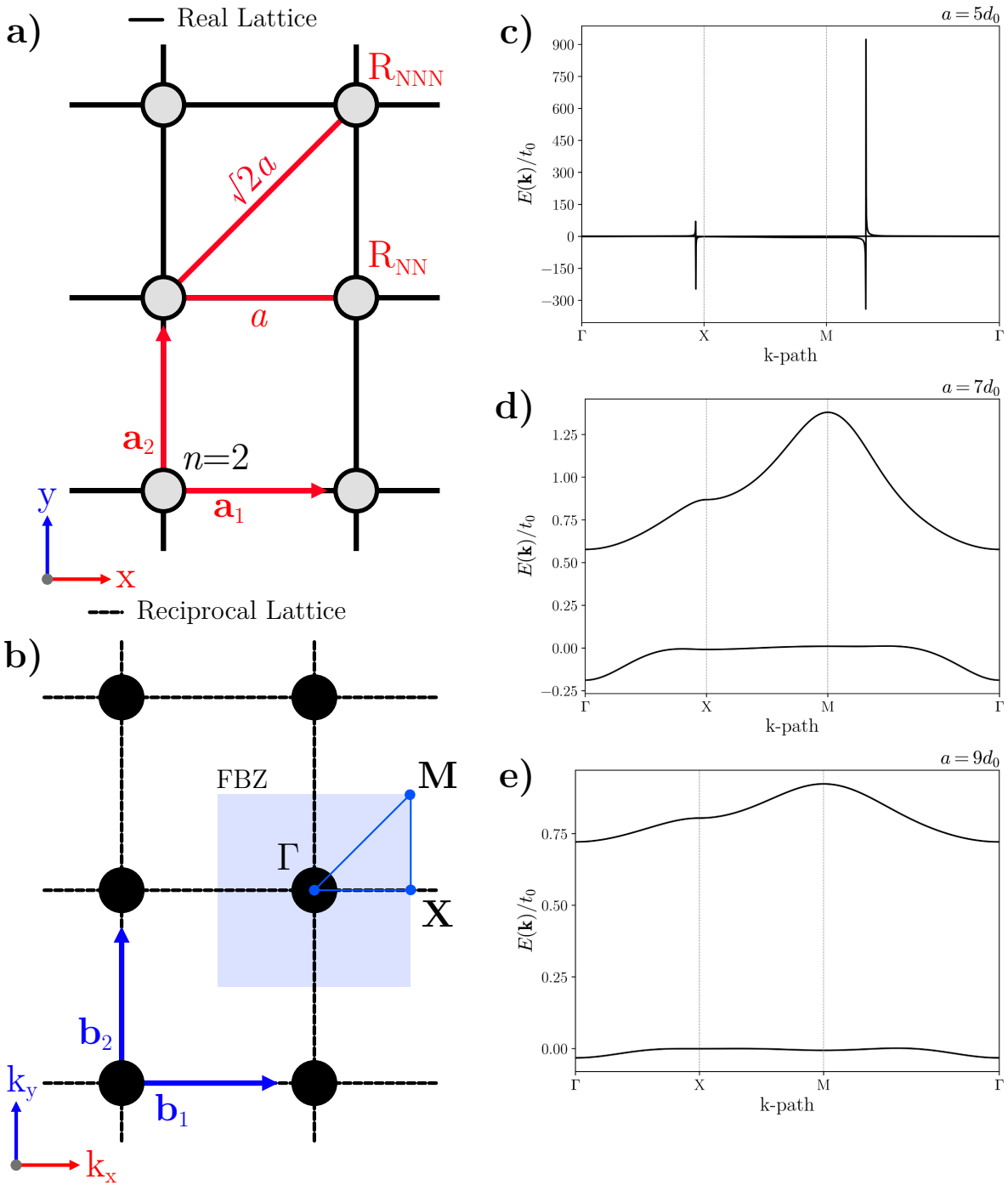


Figure 1: (a) Real-space square lattice of crystal A, indicating the lattice vectors and nearest-neighbor distances. (b) Corresponding reciprocal lattice, first Brillouin zone, and high-symmetry points. Electronic band structures along $\Gamma - X - M - \Gamma$ for (c) $a = 5d_0$, (d) $a = 7d_0$, and (e) $a = 9d_0$.

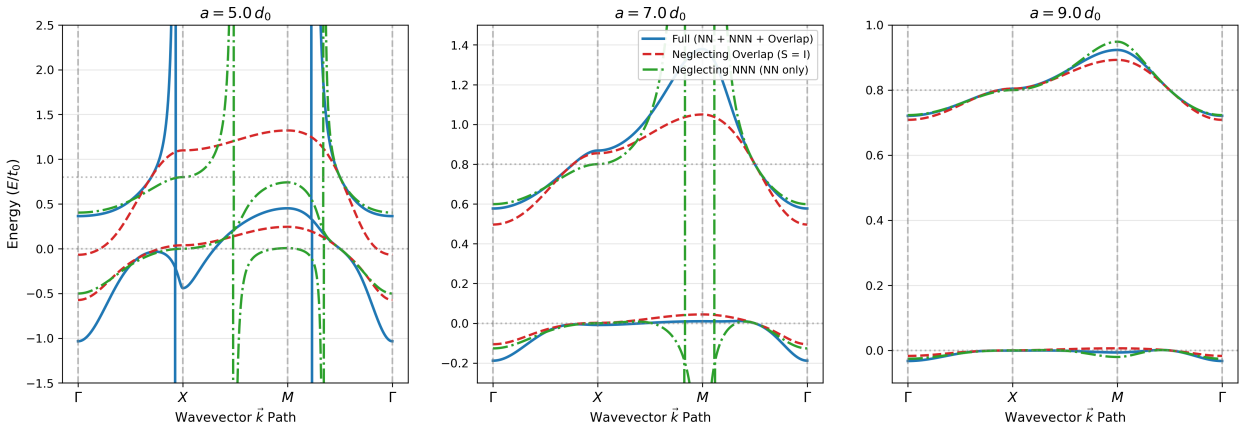


Figure 2: Electronic band structure along high-symmetry points for the square lattice showing the "Full" model with NN + NNN + Overlap (blue line), the overlap truncation with $\bar{\eta}(\mathbf{k}) = \mathbf{I}$ (red dashed line), and the inclusion of only nearest neighbors (green dashed line) for lattice parameters $a/d_0 = 5.0, 7.0, 9.0$ (left, middle, and right, respectively).

Considering first- and second-nearest-neighbor interactions, the overlap and hopping matrices are

$$\bar{\eta}(\mathbf{k}) = \begin{pmatrix} 1+\eta_{11}(a)f_{\mathrm{NNN}} & \eta_{12}(a)f_{\mathrm{NNN}} & \eta_{11}(d)f_{\mathrm{NN}} \\ \eta_{12}(a)f_{\mathrm{NNN}} & 1+\eta_{22}(a)f_{\mathrm{NNN}} & \eta_{12}(d)f_{\mathrm{NN}} \\ \eta_{11}(d)f_{\mathrm{NN}} & \eta_{12}(d)f_{\mathrm{NN}} & 1+\eta_{11}(a)f_{\mathrm{NNN}} \end{pmatrix}, \tag{8}$$

and

$$\bar{t}(\mathbf{k}) = \begin{pmatrix} t_{11}(a)f_{\mathrm{NNN}} & t_{12}(a)f_{\mathrm{NNN}} & t_{11}(d)f_{\mathrm{NN}} \\ t_{12}(a)f_{\mathrm{NNN}} & t_{22}(a)f_{\mathrm{NNN}} & t_{12}(d)f_{\mathrm{NN}} \\ t_{11}(d)f_{\mathrm{NN}} & t_{12}(d)f_{\mathrm{NN}} & t_{11}(a)f_{\mathrm{NNN}} \end{pmatrix}, \tag{9}$$

where $d = \sqrt{13}a/4$ is the distance between the two sites in the basis. The corresponding nearest-neighbor (NN) and next-nearest-neighbor (NNN) structure factors are

$$f_{\mathrm{NN}}(\mathbf{k}) = 4\cos\left(\frac{k_x a}{2}\right)\cos\left(\frac{k_y b}{2}\right), \tag{10}$$

$$f_{\mathrm{NNN}}(\mathbf{k}) = 2\cos(k_x a), \tag{11}$$

with $b = 1.5a$. Unlike crystal A, the nearest-neighbor contributions connect different atomic sites, whereas the next-nearest-neighbor terms describe interactions along the horizontal lattice direction.

The rectangular lattice and its corresponding reciprocal lattice are illustrated in Figs. 3(a) and (b), respectively. The electronic band structures were calculated along the high-symmetry path $\Gamma - X - M - Y - \Gamma$ for lattice constants $a = 5d_0$, $7d_0$, and $9d_0$.

For $a = 5d_0$ [Fig. 3(c)], the electronic bands exhibit pronounced divergences, similar to those observed for crystal A. These features are associated with the singular behavior of the overlap matrix. Consequently, the divergent solutions cannot be interpreted as physically meaningful electronic energies within the present approximation. Additionally, we note that only neglecting the overlap integrals removes the divergences seen in this limit [Fig. 4]. This behavior suggests that at small length scales, the orbitals contributing to these overlaps lose their linear independence and produce singular overlap matrices.

For $a = 7d_0$ [Figs. 3(d) and 4], the three electronic bands become well-defined. The upper band remains separated from the two lower bands, while the latter approach each other near the high-symmetry points X and M. This behavior reflects the interplay between the different orbital energies and the hopping interactions connecting the two atomic sites.

Increasing the lattice constant to $a = 9d_0$ [Fig. 3(e)] further reduces the band dispersion. As in crystal A, the exponential decay of the hopping and overlap amplitudes suppresses the momentum-dependent contributions to the generalized eigenvalue problem. Consequently, the bands progressively approach the isolated-orbital energies, with two bands tending toward $\epsilon_1 = 0$ and the remaining band approaching $\epsilon_2 = 0.8t_0$.

The presence of three bands, rather than two, is a direct consequence of the additional atomic site in the primitive cell. In the limit of large interatomic separation, the two orbitals with on-site energy ϵ_1 become degenerate, while the third band remains separated by the on-site energy difference $\epsilon_2 - \epsilon_1$.

References

Girvin, S. M., and Yang, K., Modern Condensed Matter Physics (Cambridge University Press, Cambridge, 2019).

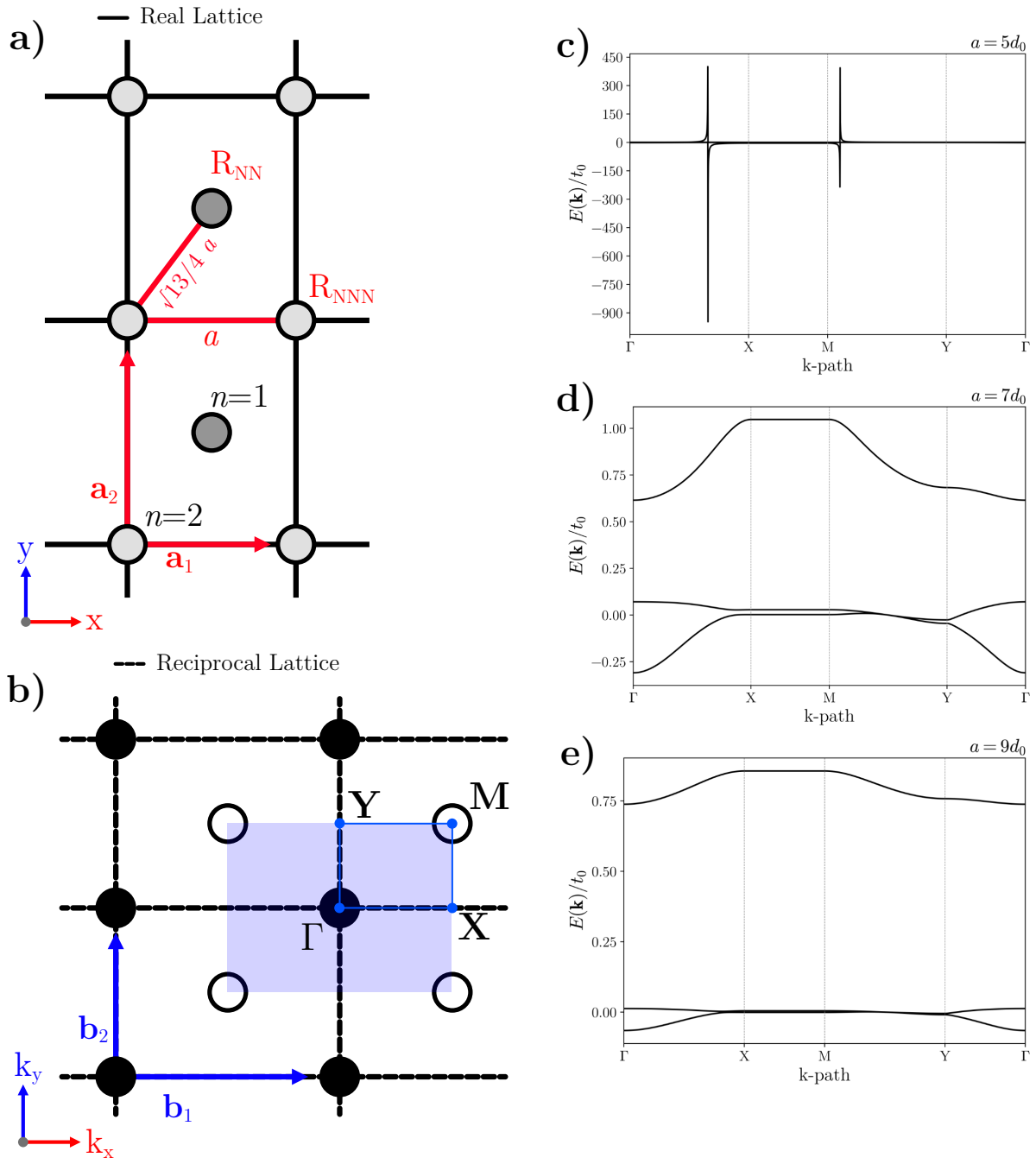


Figure 3: (a) Real-space rectangular lattice of crystal B, showing the two atomic sites, primitive lattice vectors, and nearest-neighbor distances. (b) Corresponding reciprocal lattice, first Brillouin zone, and high-symmetry points. Electronic band structures along $\Gamma - X - M - Y - \Gamma$ for (c) $a = 5d_0$, (d) $a = 7d_0$, and (e) $a = 9d_0$.

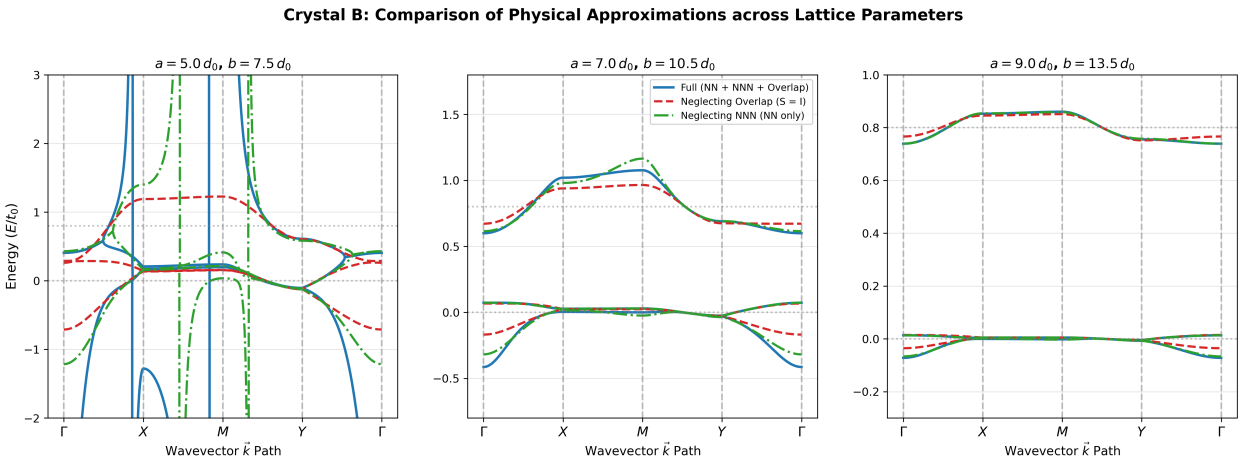


Figure 4: Electronic band structure along high-symmetry points for the rectangular lattice showing the "Full" model with NN + NNN + Overlap (blue line), the overlap truncation with $\bar{\eta}(\mathbf{k}) = \mathbf{I}$ (red dashed line), and the inclusion of only nearest neighbors (green dashed line) for lattice parameters $a/d_0 = 5.0, 7.0, 9.0$ (left, middle, and right, respectively).